

TRUSTIA AI — MASTER INVESTOR PITCH DECK (2026)

Turnkey SAE Level-4 Autonomous Driving Retrofit Kit & Deterministic Autonomy Stack for Commercial EVs

Founder & Chief Architect: Murat Furkan Bayram (Age 17, 80% Equity) Hardware & Integration Lead: Denizcan Ozcan (ASELSAN Candidate Eng., TEKNOFEST Finalist)	The Pre-Seed Ask: \$500,000 USD on a \$5M Post-Money Cap SAFE Core Entity & Incubation: Trustia AI (ITO BTM Fulya Campus Resident)	Engineering Maturity: 16,000 LOC C++/Python, 1,301/1,301 Tests (100%) Global Track Record: Dubai RTA \$1.2M Challenge Official Entry (Submitted)
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1. THE \$118B PROBLEM: BESPOKE ROBOTAXIS ARE TOO EXPENSIVE TO SCALE

The Flaw in Waymo & Cruise: Industry giants spend \$250,000 to \$350,000+ manufacturing bespoke autonomous pods from scratch. Because of massive capital expenditure (CAPEX) and punishing vehicle depreciation, driverless fleets cannot scale commercially to emerging markets, taxi operators, or campus shuttles.

Market Urgency: The global autonomous vehicle market will reach \$118 Billion by 2030. Major municipalities (such as Dubai's legal mandate for 25% driverless transport by 2030) urgently need low-cost, rapidly deployable autonomous solutions.

2. THE TRUSTIA SOLUTION: \$35,000 MODULAR RETROFIT IN 48 HOURS

Trustia AI turns mass-produced electric passenger vehicles (specifically the Hyundai Ioniq 5 E-GMP platform) into commercial SAE Level-4 Robotaxis in under 48 hours. By interfacing directly with the vehicle's CAN-FD drive-by-wire system without cutting or modifying the structural chassis, Trustia delivers a 70% CAPEX reduction and a full 14-month payback period for fleet operators.

3. PROPRIETARY TECHNOLOGY: 16,000 LOC DETERMINISTIC CORE (ZERO BLACK BOX)

While traditional end-to-end neural nets suffer from hallucination and unpredictability, Trustia employs a mathematically provable, deterministic architecture:

- Kinematic Hybrid A* Path Planning: Solves non-holonomic vehicle dynamics (Ackermann steering constraints, curvature, slip angles) in under 40ms.
- GPS-Denied Centimeter SLAM: 400Hz Error-State Kalman Filter (ESKF) fused with 3D Normal Distributions Transform (NDT) LiDAR SLAM provides 5cm localization in urban canyons, tunnels, and GPS-jammed zones.
- 1,301 Automated Tests: Rigorous CI/CD test suite spanning unit algorithms, hardware-in-the-loop (HIL) simulators, and CAN-FD fault injections at 100% pass rate.
- ISO 26262 ASIL-D Safety: Hardware E-Stop and automated Minimal Risk Maneuver (MRM) ensures safe curbside pulling in case of sensor degradation.

4. PRODUCTION-READY SENSOR & COMPUTE SUITE

Subsystem	Component / Model	Key Specifications & Function
Roof 3D LiDAR	Ouster OS2-128 Rev 7	128 laser channels, 240m range, 2.62M pts/sec, 360° primary point cloud
Blind Spot LiDAR (2x)	Livox Mid-360	360°x59° FOV, front/rear bumper zero blind-spot curb & pedestrian tracking
Long-Range Radars (2x)	Continental ARS 408-21 (77GHz)	250m range, FMCW all-weather penetration (heavy fog, dust storms, desert heat)
Perception Cameras (4x)	Leopard Sony IMX390 GMSL2	120dB HDR automotive grade, LED flicker mitigation, 360° vision
Central AI Compute	NVIDIA Jetson AGX Orin 64GB	275 TOPS INT8, Saeed J501 industrial carrier, 100Hz real-time deterministic loop
Drive-by-Wire Bridge	Kvaser U100 CAN-FD	5 Mbps high-speed CAN-FD, galvanic isolation, LKAS/SCC steering & brake injection

5. UNIT ECONOMICS & DUAL-LAYER REVENUE MODEL

1. Hardware Kit Sales (Upfront CAPEX): \$35,000 per retrofit kit (35% gross profit margin).
 2. Autonomy-as-a-Service (AaaS Recurring ARR): \$0.18/km or \$450/month per vehicle for tele-operations, continuous OTA updates, and HD map syncing.
- Fleet ROI Impact: Each retrofitted vehicle generates \$55,500/year in net operating savings by eliminating driver shift costs, achieving full capital payback in 14 months.

6. THE INVESTMENT ASK: \$500,000 PRE-SEED

We are raising \$500,000 USD on a \$5M Post-Money Cap SAFE to fund our 18-month operational runway:

- 45% Hardware & Sensor Procurement: Full conversion of our first 2 Hyundai Ioniq 5 proving ground test vehicles.
- 35% Core Engineering Talent: Embedded systems, SLAM, and ROS2 engineers.
- 15% Proving Ground & Track Testing: Live validation on closed courses in Silicon Valley and Dubai RTA proving grounds.
- 5% ISO 26262 ASIL-D Certification & IP: Global patent filings and safety audits.

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